Ch 5

METABOLISM OF HEME

1 Major sites of heme biosynthesis are:

a) liver and kidney
b) liver and bone marrow
c) RBCs and kidney
d) Heart and kidney

(Department book)

2 One of the following is true about δ-aminolevulinate synthase (ALA synthase):

a) It is cytosolic enzyme
b) The enzyme is allosterically inhibited by ALA
c) The enzyme needs pyridoxal phosphate (PLP) as a coenzyme
d) The enzyme catalyzes condensation of Alanine and succinyl-CoA

(Department book)

3 Neuropsychiatric manifestations of porphyria are characterized by the accumulation of:

a) δ-aminolevulinate or porphobilinogen
b) uroporphyrinogen III
c) coproporphyrinogen III
d) protoporphyrinogen

(Department book)

4 Acute intermittent porphyria is due to deficiency of the following enzyme:

a) Uroporphyrinogen III synthase
b) Uroporphyrinogen decarboxylase
c) Uroporphyrinogen I synthase
d) Ferrochelatase

(Department book)

5 Porphyria cutanea tarda is due to deficiency of the following enzyme:

a) Uroporphyrinogen III synthase
b) Uroporphyrinogen decarboxylase
c) Uroporphyrinogen I synthase
d) Ferrochelatase

(Department book)

6 Effect of lead poisoning on heme metabolism is:

a) Interfere with OH containing enzyme in heme biosynthesis
b) Inhibition of ALA dehydratase and ferrochelatase
c) Elevated levels of uroporphyrin in urine.
d) Can lead to megaloblastic anemia

(Department book)

7 Which of the following organs is NOT involved in heme catabolism:

a) Liver
b) Spleen
c) Kidney
d) Intestine

(Department book)

8 In the liver, bilirubin is conjugated with:

a) Glucuronic acid
b) Hyaluronic acid
c) Galacturonic acid
d) Arachidonic acid

(Department book)

9 The rate-limiting step in bilirubin metabolism is:

a) Degradation of heme is by the heme oxygenase system
b) Conjugation of bile by glucuronyltransferase
c) Reduction of bilirubin to biliverdin, by biliverdin reductase
d) Secretion of conjugated bilirubin into the bile

(Department book)

10 Conjugated bilirubin is characterized by:

a) Water soluble
b) Bound to plasma protein
c) Can cross blood brain barrier
d) Increased in hemolytic anemia

(Department book)

11 Which of the following is not a characteristic of unconjugated bilirubin:

a) Water insoluble
b) Is not bound to plasma proteins
c) Normal blood level ranges between 0.2-0.9 mg/dL
d) If increased > 20 mg/dl can lead to kernicterus

(Department book)

12 Crigler-Najjar Syndrome, type I is characterized by:

a) Partial deficiency of UDP-glucuronyltransferase
b) Serum unconjugated bilirubin does not exceed 20 mg/dL
c) autosomal dominant disease
d) Patients do not respond to phenobarbital therapy

(Department book)

METABOLISM OF HEME

13 ..... is condition caused by a defect in the uptake of unconjugated bilirubin by the liver:

a) Crigler-Najjar Syndrome, type I
b) Crigler-Najjar Syndrome, type II
c) Gilbert Syndrome
d) Rotor syndrome

(Department book)

14 Which of the following is a cause of conjugated hyperbilirubinemia:

a) Crigler-Najjar Syndrome, type I
b) Crigler-Najjar Syndrome, type II
c) Gilbert Syndrome
d) Obstructive jaundice

(Department book)

15 Which of the following is NOT a characteristic of physiologic neonatal jaundice:

a) Partly results from increased hemolysis
b) Partly results from immaturity of the enzyme UDP-glucuronyltransferase.
c) Serum bilirubin usually exceeds 13 mg/dL
d) Can be treated by Phenobarbital

(Department book)

16 Kernicterus is characterized by:

a) Can be caused by Crigler-Najjar Syndrome, type I
b) Conjugated hyperbilirubinemia
c) Can be caused by Gilbert syndrome
d) Can be caused by Dubin-Johnson syndrome

(Department book)

17 Conjugated hyperbilirubinemia is characterized by:

a) Acholuric urine
b) Stercobilin in the feces usually decreases
c) Urobilin in the urine usually increases
d) Normal liver function tests

(Department book)

18 Which serum enzyme elevation is most diagnostic of obstructive jaundice?

a) Alanine amino transferase (ALT)
b) Aspartate amino transferase (AST)
c) Lactate dehydrogenase (LDH)
d) Alkaline phosphatase (ALP)

(Department book)

19 A rise in serum direct bilirubin would be expected in:

a) Hemolytic jaundice
b) Absence of glucuronyl transferase as in the newborn
c) Decreased hepatic uptake of bilirubin (Gilbert's Disease)
d) Biliary obstruction

(Department book)

20 Fecal stercobilinogen is increased in:

a) Hemolytic anemia
b) Hepatic jaundice
c) Viral hepatitis
d) Obstructive jaundice

(Department book)

21 In hemolytic jaundice, urine shows:

a) Absence of bilirubin and urobilinogen
b) Presence of bilirubin and urobilinogen.
c) Absence of bilirubin and presence of urobilinogen.
d) Presence of bilirubin and absence of urobilinogen

(Department book)

22 Heme biosynthesis occurs in:

a. Spleen and liver
b. Liver and brain
c. Liver and bone marrow
d. Spleen and brain

(Department book)

23 High levels of lead can affect heme metabolism by inhibition of:

a. ALA synthase
b. Ferrochelatase
c. ALA decarboxylase
d. Coproporphyrinogen oxidase

(Department book)

24 In which of the following porphyriasphotosensitivity is not present?

a. Acute intermittent porphyria
b. Porphyria cutanea tarda
c. Congenital erythropoietic porphyria
d. Protoporphyria

(Department book)

METABOLISM OF HEME

25 Photosensitivity manifestations in porphyria disease are due to:

a. decrease ATPase activity in nerve cells.
b. accumulation of ALA or PBG
c. release of free radicals due to activation of porphyrins
d. vitamin B6 deficiency

(Department book)

26 Unconjugated bilirubin is elevated in which of the following conditions?

a. Dubin-Johnson Syndrome.
b. Physiologic neonatal jaundice.
c. Rotor syndrome.
d Obstruction of the biliary passages.

(Department book)

27 A jaundiced patient has elevated serum levels of ALT, AST and γ-GT. This patient most probably has:

a. hemolytic jaundice
b. obstructive jaundice
c. hepatocellular jaundice
d. acholuric jaundice

(Department book)

28 Acholuric jaundice may occur in which of the following conditions?

a. Viral hepatitis
b. Favism
c. Rotor syndrome
d. Dubin-Johnson syndrome

(Department book)

29 Pyridoxal phosphate is necessary for heme synthesis. Which out of the following enzymes requires pyridoxal phosphate as a coenzyme?

a. ALA synthase
b. ALA dehydratase
c. Ferrochelatase
d. PBG deaminase

(Department book)

30 Which serum enzyme elevation is most diagnostic of obstructive jaundice?

a. Alanine amino transferase (ALT)
b. Alkaline phosphatase (ALP)
c. Aspartate amino transferase (AST)
d. Lactate dehydrogenase (LDH)

(Department book)

31 During heme synthesis, which of the following enzymes are inhibited by lead poisoning?

a) ALA synthase and Protoporphyrin oxidase
b) ALA synthase and Ferrochelatase
c) ALA dehydratase and Protoporphyrin oxidase
d) ALA dehydratase and Ferrochelatase

(pervious exam)

32 A patient with jaundice has high total bilirubin that is mainly unconjugated, high concentration of sterzobilirin the feces. The level of conjugated bilirubin in the blood plasma is normal. What type of jaundice can be suspected in this patient?

a) Hemolytic anemia
b) Hepatocellular jaundice
c) Obstructive Jaundice
d) Rotor syndrome

(pervious exam)

33 What is a key characteristic of conjugated bilirubin in comparison to unconjugated bilirubin?

a) It is transported in the blood attached to albumin
b) It is water soluble and can appear in urine
c) is fat-soluble and cannot be excreted in urine
d) accumulates in the bloodstream during hemolysis

(pervious exam)

34 Photosensitivity manifestations in porphyria disease are due to:

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(pervious exam)

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(pervious exam)

METABOLISM OF HEME

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c. hepatocellular jaundice
d. acholuric jaundice

(pervious exam)

37 Heme biosynthesis occurs in:

a. Spleen and liver
b. Liver and brain
c. Liver and bone marrow
d. Spleen and brain

(pervious exam)

38 High levels of lead can affect heme metabolism by inhibition of:

a. ALA synthase
b. Ferrochelatase
c. ALA decarboxylase
d. Corpoporphyrinogen oxidase

(pervious exam)

39 In which of the following porphyrias photosensitivity is not present?

a. Acute intermittent porphyria
b. Porphyria cutanea tarda
c. Congenital erythropoietic porphyria
d. Protoporphyria

(pervious exam)

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c. Ferrochelatase
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(pervious exam)

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b. Alkaline phosphatase (ALP)
c. Aspartate amino transferase (AST)
d. Lactate dehydrogenase (LDH)

(pervious exam)

42 118- Acholuric jaundice may occur in which of the following conditions?

a. Viral hepatitis
b. Favism
c. Rotor syndrome
d. Dubin-Johnson syndrome

(pervious exam)

Answers

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
b	c	a	c	b	b	c	a	d	a	b	d	c	d	c
16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
a	b	d	d	a	c	c	b	a	c	b	c	b	a	b
31	32	33	34	35	36	37	38	39	40	41	42			
d	a	b	c	b	c	c	b	a	a	b	b			

Ch 9

PURINES & PYRIMIDINES

1 In de novo synthesis of purine nucleotides, the donors of nitrogen atoms are:

a) Aspartate and glutamate
b) Aspartate and glycine
c) Aspartate, glutamate, and glycine
d) Aspartate, glutamine, and glycine

(Department book)

2 ..... is the committed step in de nonvo purine nucleotide:

a) PRPP synthetase
b) Glutamine:PRPP amidotransferase
c) Hypoxanthine-guanine phosphoribosyl transferase (HGPRT)
d) Adenosine kinase

(Department book)

3 Which of the following cells uses salvage of free bases as its major form of nucleotide generation?

a) Nephron cell
b) RBCs
c) Hepatocyte
d) Osteoclasts

(Department book)

4 Which one of the following enzymes is important for free purine salvage system?

a) PRPP synthetase
b) Glutamine:PRPP amidotransferase
c) Hypoxanthine-guanine phosphoribosyl transferase (HGPRT)
d) Adenosine kinase

(Department book)

5 Salvage of Hypoxanthine to IMP is catalyzed by:

a) Adenosine phosphoribosyl transferase
b) Deoxycytidine kinase
c) Hypoxanthine-guanine phosphoribosyl transferase
d) Adenylosuccinate synthetase

(Department book)

6 Free purine bases which can be salvaged are:

a) Adenine and guanine
b) Adenine and hypoxanthine
c) Guanine and hypoxanthine
d) Adenine, guanine, and hypoxanthine

(Department book)

7 One of the following does not produce feedback inhibition of PRPP synthetase:

a) AMP
b) ADP
c) GMP
d) GTP

(Department book)

8 An allosteric inhibitor of adenylosuccinate synthetase is:

a) AMP
b) GMP
c) ADP
d) GDP

(Department book)

9 ..... is an allosteric inhibitor of IMP-dehydrogenase.

a) AMP
b) ADP
c) GMP
d) GDP

(Department book)

10 ..... is the end product of purine metabolism in human is:

a) Inosine
b) Urea
c) Uric acid
d) Xanthine

(Department book)

11 Plasma uric acid level in males is ....., whereas its level in females is .....

a) 4-7, 1-3 mg/dL respectively
b) 3-6, 8-10 mg/dL respectively
c) 2-4, 4-7 mg/dL respectively
d) 4-7, 3-6 mg/dL respectively

(Department book)

12 One of the following disorders cannot be considered a cause of gout:

a) Congenital renal disease.
b) Lesch-Nyhan Syndrome
c) Xanthine oxidase deficiency
d) Von Gierke's disease

(Department book)

PURINES & PYRIMIDINES

13 Which gene is defective in patients with Lesch-Nyhan syndrome?

a) PRPP synthetase
b) Adenosine deaminase
c) Purine nucleoside phosphorylase
d) Hypoxanthine-guanine phosphoribosyl transferase

(Department book)

14 Which one of the following enzymes can be inhibited by Allopurinol?

a) Glucose 6-phosphatase
b) Adenosine deaminase
c) Xanthine oxidase
d) Purine nucleoside phosphorylase

(Department book)

15 Allopurinol can be used to treat gout because of its ability to inhibit which one of the following reactions?

a) AMP to XMP
b) Xanthine to uric acid
c) Inosine to hypoxanthine
d) IMP to XMP

(Department book)

16 Hypourecemia can occur in:

a) Xanthine oxidase deficiency
b) Congenital renal disease
c) Leukemia
d) Von Gierke's disease

(Department book)

17 Immune dysfunctions are associated with a defective of:

a) PRPP synthetase
b) Glutamine:PRPP amidotransferase
c) Both adenosine deaminase and purine nucleoside phosphorylase
d) Hypoxanthine-guanine phosphoribosyl transferase

(Department book)

18 Allopurinol can be used to treat gout because of it inhibits:

a. Glucose 6-phosphatase
b. Adenosine deaminase
c. Xanthine oxidase
d. Purine nucleoside phosphorylase

(Department book)

19 One of the following is not a cause of gout:

a. Xanthine oxidase deficiency
b. Lesch-Nyhan Syndrome
c. Congenital renal disease
d. Von Gierke's disease

(Department book)

20 As regards gout, which of the following is incorrect?

a. Alcohol intake may precipitate an acute attack of gouty arthritis
b. It is treated by eating of nucleoprotein rich diet (cellular proteins) e.g.meat, liver and kidney
c. It may be caused by diseases like leukemia and cancer
d. It is treated by uricosuric drugs

(Department book)

21 Severe combined immunodeficiency disease may be caused by deficiency of which of the following enzymes?

a. Glucose-6-phosphatase
b. Xanthine oxidase
c. PRPP synthetase
d. Adenosine deaminase

(Department book)

22 A drug which decreases uric acid synthesis by inhibiting the enzyme xanthine oxidase is:

a. allopurinol
b. methotrexate
c. colchicine
d. probenecid

(Department book)

23 A 2-year-old child with a mental retardation and manifestations of self-mutilation has a high concentration of uric acid. Diagnosis revealed HGPRTase deficiency. What is the most likely diagnosis?

a. Lesch-nyhansyndrome
b. Von gierkedisease
c. Hartnupdisease
d. Gilbert disease

(Department book)

PURINES & PYRIMIDINES

24 Which of the following is the coenzyme required for the enzyme ribonucleotide reductase?

a. Glutathione
b. Thioredoxin
c. NADPH
d. FADH

(Department book)

25 Which of the following enzymes is defective in patients with Lesch-Nyhan syndrome?

a. PRPP synthetase
b. Adenosine deaminase
c. Purine nucleoside phosphorylase
d. Hypoxanthine-guanine phosphoryl transferase

(pervious exam)

26 Which of the following is the coenzyme required for the enzyme ribonucleotide reductase?

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(pervious exam)

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(pervious exam)

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a. Lesch-nyhan syndrome
b. Von gierke disease
c. Hartnup disease
d. Gilbert disease

(pervious exam)

30 A 50-year-old patient with hyperuricemia is diagnosed with gout. The metabolism of which of the following is disturbed in this patient:

a) Pyrmidines
b) Amino acids
c) Purines
d) Fats

(pervious exam)

31 Allopurinol can be used to treat gout because of it inhibits:

a. Glucose 6-phosphatase
b. Adenosine deaminase
c. Xanthine oxidase
d. Purine nucleoside phosphorylase

(pervious exam)

32 One of the following is not a cause of gout:

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b. Lesch-Nyhan Syndrome
c. Congenital renal disease
d. Von Gierke's disease

(pervious exam)

Answers

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a	c	c	a	b	d	a	a	b	d	b	d	a	a	c
31	32													
c	a													